

Modular SOC Decomposition Framework

Architecture Documentation for Multi-Model Intercomparison

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2026-02-12

1. Overview

This document defines a modular architecture for soil organic carbon (SOC) decomposition modelling within the HIKET multi-model intercomparison framework. The goal is to define all models — Yasso07, Yasso15, Yasso20, RothC, and Q-model — through common functional interfaces that cleanly separates **litter input processing**, **environmental modification of kinetics**, and **decomposition dynamics**.

This separation enables modularity.

2. General Framework

2.1 Core Equation

The simulated soil organic carbon over time is expressed as:

$$\mathbf{C}(t, k) = f_{d_k} \left(f_{i_k}(\mathbf{I}(t)), \xi_k(t) \right)$$

where:

Symbol	Description	Dimensions
$\mathbf{C}(t)$	SOC pool vector at time t	$[n_{\text{pools}} \times 1]$
$\mathbf{I}(t)$	Raw litter input data at time t	model-agnostic format
$f_i(\cdot)$	Input compatibility layer — transforms raw inputs into model-specific pool inputs	$\mathbf{I}(t) \rightarrow \mathbf{J}(t)$
$f_d(\cdot)$	Decomposition function — the structural core of each model	$(\mathbf{J}(t), \xi(t)) \rightarrow \mathbf{C}(t)$
$\xi(t)$	Kinetic modifier — composite environmental scalar(s)	$\in \mathbb{R}^{n_\xi}$

The k denotes model-specific functions.

The kinetic modifier is itself composed of independent environmental response functions:

$$\xi(t) = \xi_T(T(t)) \cdot \xi_M(M(t)) \cdot \xi_e$$

where ξ_T is the temperature modifier, ξ_M is the moisture/precipitation modifier, and ξ_e represents any additional static or time-varying modifiers, likely edaphic (e.g., clay content, soil cover).

2.2 Architecture Schema

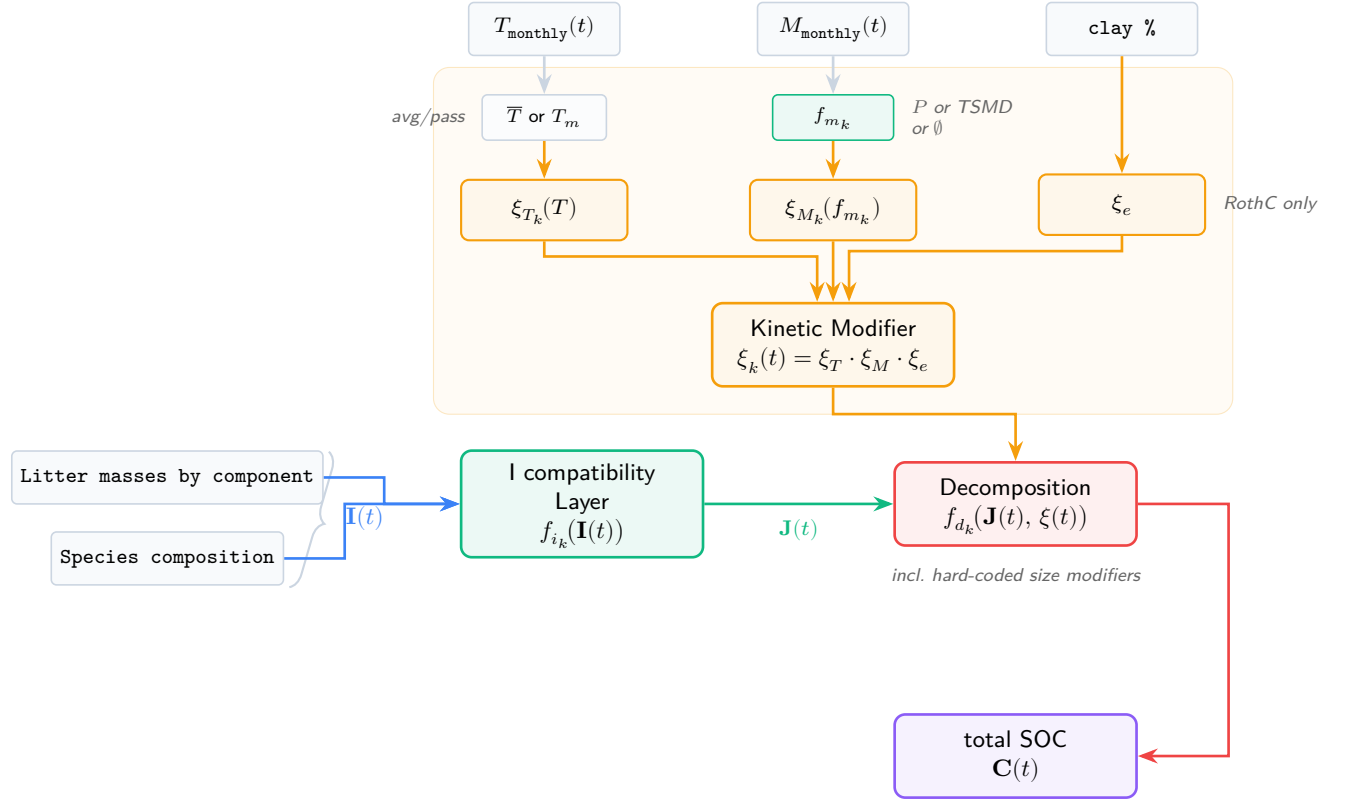


Figure 1: Modular architecture of the SOC decomposition framework. Raw inputs $\mathbf{I}(t)$ are transformed by the input compatibility layer f_i into model-specific inputs $\mathbf{J}(t)$. Environmental drivers are combined into the composite kinetic modifier $\xi(t)$. Both feed into the model-specific decomposition function f_d .

3. Input requirements

3.1 Standardised input format — $\mathbf{I}(t)$

Dimensions $N_{plots} \times N_{timesteps}$

The standard input data for all models $\mathbf{I}(t)$ contains litter mass inputs from six cohorts (foliage, fine roots, branches, coarse roots, stump, stem). The species group (pine, spruce, birch) are partitioned based on species composition data, which are assumed constant per plot.

3.2 Standardised environmental drivers

3.2.1 Time-variant drivers

Dimensions $N_{plots} \times N_{timesteps}$

Temperature and moisture inputs are provided as monthly time series. Temperature relies only on a compatibility layer to aggregate to annual summaries where needed. Moisture is more complicated since the models rely on different inputs (precipitation vs topsoil moisture deficit), so we need a compatibility layer to convert precipitation to soil moisture deficit for RothC, and to pass precipitation directly to Yasso. The Q-model does not explicitly include a moisture modifier, but implicitly via litter quality.

3.2.2 Static edaphic modifiers

Dimensions $N_{plots} \times 1$

Clay content is provided as a scalar per plot, as well as tree species composition.

3. Output structure

3.1 Model-specific output

Dimensions $N_{plots} \times N_{timesteps} \times N_{pools}$, replicated for each model (different data frames for each model, with different pool names and numbers).

3.1 Total SOC output

Dimensions $N_{plots} \times N_{timesteps} \times N_{models}$

3. Components requirements

3.1 Input Compatibility Layer — $f_i(\mathbf{I}(t))$

The input layer f_i maps this standardised input into the model-specific input vector $\mathbf{J}(t)$:

$$f_i : \mathbf{I}(t) \longrightarrow \mathbf{J}(t) \in \mathbb{R}^{n_{\text{input pools}}}$$

The mapping differs across models:

Model	Input pools $\mathbf{J}(t)$	Key transformation
Yasso07/15/20	AWEN fractions (A, W, E, N)	Total mass to AWEN via component-specific coefficients
RothC	DPM, RPM, BIO, HUM, IOM	Total mass $\rightarrow$ DPM/RPM split via component-dependent coefficients

Model	Input pools $\mathbf{J}(t)$	Key transformation
Q-model	Quality-distributed cohorts per tissue type	Separate cohorts retained; quality q_0 assigned per species and component

3.2 Kinetic Modifier — $\xi(t)$

The kinetic modifier captures all environmental effects on decomposition rates. It is composed multiplicatively:

$$\xi(t) = \xi_T(T(t)) \times \xi_M(f_{m_k}(M(t))) \times \xi_k$$

Each sub-modifier is a vector over time (or, for edaphic effects ξ_k , a scalar, constant over time) that rescales the baseline decomposition rates.

3.2.1 Temperature Modifier — ξ_T

All functions accept the same input, with the only compatibility layer being a monthly to annual averaging.

The functional form of the temperature response differs across models:

Model	Form	Expression
Yasso07/15	Gaussian-like, 4-point annual	$\xi_T = \frac{1}{4} \sum_{i=1}^4 \exp(\beta_1 T_i + \beta_2 T_i^2)$, T_i from annual mean + amplitude
Yasso20	Gaussian-like, exact monthly	$\xi_T = \frac{1}{12} \sum_{m=1}^{12} \exp(\beta_1 T_m + \beta_2 T_m^2)$, T_m are observed monthly values
RothC	Arrhenius-based	$\xi_T = \frac{47.9}{1 + \exp(\frac{106}{T+18.3})}$
Q-model	Linear	$\xi_T = \max(0, a_T + b_T \cdot T)$

3.2.2 Moisture/Precipitation Modifier — ξ_M

Moisture functions of different models accept different inputs, so we need another compatibility layer, f_{m_k} . Yasso uses precipitation, RothC uses topsoil moisture deficit, and Q-model does not explicitly include a moisture modifier (but implicitly via litter quality).

Model	Driver	Notes
Yasso07/15	Annual precipitation P	$\xi_M = 1 - \exp(\gamma \cdot P/1000)$; pool-group-specific for Yasso15
Yasso20	Monthly precipitation (summed annually)	Same formula; P derived internally as annual sum of monthly values
RothC	Topsoil moisture deficit	Rate-modifying factor from TSMD accumulation
Q-model	—	Not explicitly included (implicitly via litter quality)

3.2.3 Additional Static Modifiers — ξ_k

Modifier	Applicable models	Description
Clay (ξ_{clay})	RothC	Modifies BIO/HUM partitioning: $x = 1.67(1.85 + 1.60 \exp(-0.0786 \% \text{clay}))$
Soil cover	RothC	Binary modifier on decomposition rate for bare soil, WON'T BE IMPLEMENTED!

3.2.4 Hard-coded Static Modifiers

Both Q and Yasso have static modifiers dependent on size that are specific to litter classes. These are hard-coded in each model function, rather than passed as separate modifiers.

For the Yasso family of models, all size modifiers are hard-coded in the R wrappers

4. Decomposition Equations

This section presents the complete decomposition dynamics for each model

4.1 Yasso07

Pool structure

Five pools: $\mathbf{C} = [C_A, C_W, C_E, C_N, C_H]^T$, representing Acid-soluble, Water-soluble, Ethanol-soluble, Non-soluble (lignin-like), and Humus.

Dynamics

$$\frac{d\mathbf{C}}{dt} = \mathbf{A} \mathbf{C} + \mathbf{I}$$

where $\mathbf{I} = [I_A, I_W, I_E, I_N, 0]^T$ (no direct input to Humus).

Diagonal elements (loss from pool i):

$$A_{ii} = -\alpha_i \cdot \xi_T \cdot \xi_M$$

where α_i is the base decomposition rate. For the N pool, an additional size modifier applies to woody litter:

$$A_{44} = -\alpha_N \cdot \xi_T \cdot \xi_M \cdot f_{\text{size}}$$

with $f_{\text{size}} = \min(1, \exp(-\delta_1 \cdot d^{\delta_2} \cdot |r|))$, where d is the wood diameter (cm).

Off-diagonal elements (transfer from pool j to pool i):

$$A_{ij} = p_{ij} \cdot |\alpha_j \cdot \xi_T \cdot \xi_M|$$

Humus formation (row 5, columns 1–4):

$$A_{5j} = p_H \cdot |\alpha_j \cdot \xi_T \cdot \xi_M| \quad \text{for } j \in \{A, W, E, N\}$$

Respiration is the residual: $R_h = \sum_j |A_{jj}| C_j - \sum_{i \neq j} A_{ij} C_j$.

Environmental modifiers

Temperature — four-point approximation of intra-annual variation:

$$\xi_T = \frac{1}{4} \sum_{i=1}^4 \exp(\beta_1 T_i + \beta_2 T_i^2)$$

where the quarterly temperatures are derived from annual mean $\bar{T}$ and amplitude A_T :

$$\begin{aligned} T_1 &= \bar{T} + \frac{4A_T}{\pi} \left(\frac{1}{\sqrt{2}} - 1 \right), & T_2 &= \bar{T} - \frac{4A_T}{\sqrt{2}\pi} \\ T_3 &= \bar{T} + \frac{4A_T}{\pi} \left(1 - \frac{1}{\sqrt{2}} \right), & T_4 &= \bar{T} + \frac{4A_T}{\sqrt{2}\pi} \end{aligned}$$

Precipitation:

$$\xi_M = 1 - \exp\left(\gamma \cdot \frac{P}{1000}\right)$$

where P is annual precipitation (mm).

Analytical solution

For annual timestep $\Delta t = 1, \text{yr}$:

$$\mathbf{C}(t+1) = e^{\mathbf{A}} \cdot \mathbf{C}(t) + (e^{\mathbf{A}} - \mathbf{I}_5) \mathbf{A}^{-1} \mathbf{I}$$

where $e^{\mathbf{A}}$ is the matrix exponential.

Steady state:

$$\mathbf{C}^* = -\mathbf{A}^{-1} \mathbf{I}$$

Default parameters

Parameter	Value	Description
$\alpha_A, \alpha_W, \alpha_E, \alpha_N$	0.66, 4.30, 0.35, 0.22	Base decomposition rates (yr^{-1})
α_H	0.0033	Humus decomposition rate (yr^{-1})
p_H	0.04	Fraction to Humus
β_1, β_2	0.096, \$-\$0.0014	Temperature response parameters
γ	\$-\$1.21	Precipitation response parameter
δ_1, δ_2, r	\$-\$1.7, 0.86, \$-\$0.306	Size modifier parameters

4.2 Yasso15

Yasso15 retains the AWENH pool structure and transfer matrix of Yasso07, but introduces two structural changes.

Pool-group-specific climate responses. Where Yasso07 applies a single temperature and precipitation modifier to all pools, Yasso15 separates the climate response into three groups — AWE, N, and H — each with independent parameters:

$$\xi_T^{\text{AWE}} = \frac{1}{4} \sum_{i=1}^4 \exp(\beta_1 T_i + \beta_2 T_i^2), \quad \xi_T^N = \frac{1}{4} \sum_{i=1}^4 \exp(\beta_{N1} T_i + \beta_{N2} T_i^2), \quad \xi_T^H = \frac{1}{4} \sum_{i=1}^4 \exp(\beta_{H1} T_i + \beta_{H2} T_i^2)$$

Similarly for precipitation:

$$\xi_M^{\text{AWE}} = 1 - \exp(\gamma \cdot P/1000), \quad \xi_M^N = 1 - \exp(\gamma_N \cdot P/1000), \quad \xi_M^H = 1 - \exp(\gamma_H \cdot P/1000)$$

The combined climate modifier is applied per pool group when constructing the diagonal of $\mathbf{A}$:

$$\begin{aligned} A_{ii} &= -\alpha_i \cdot \xi_T^{\text{AWE}} \cdot \xi_M^{\text{AWE}} \cdot f_{\text{size}} && \text{for } i \in \{A, W, E\} \\ A_{44} &= -\alpha_N \cdot \xi_T^N \cdot \xi_M^N \cdot f_{\text{size}} \\ A_{55} &= -\alpha_H \cdot \xi_T^H \cdot \xi_M^H && \text{(no size effect)} \end{aligned}$$

Off-diagonal elements and humus formation follow the same structure as Yasso07, with transfers proportional to the source pool's modified rate.

Leaching. Yasso15 adds a precipitation-dependent leaching term to the AWEN pools (not H):

$$A_{ii} += w \cdot P/1000 \quad \text{for } i \in \{A, W, E, N\}$$

The size modifier, transfer matrix structure, analytical solution (matrix exponential), and steady-state formula ($\mathbf{C}^* = -\mathbf{A}^{-1}\mathbf{b}$) are identical to Yasso07.

4.3 Yasso20

Yasso20 shares the AWENH pool structure, pool-group-specific climate responses, and leaching of Yasso15. The structural equations are the same. The principal differences are in parameterisation and the size modifier formula.

Modified size dependence. Yasso07/15 use an exponential form:

$$f_{\text{size}}^{07/15} = \min(1, \exp(-\delta_1 \cdot d^{\delta_2} \cdot |r|))$$

Yasso20 uses a quadratic formulation:

$$f_{\text{size}}^{20} = (1 + \theta_1 d + \theta_2 d^2)^{-|r|}$$

which produces a stronger size-dependent slowdown for coarse woody debris.

Substantially different parameter values. The most consequential changes from Yasso07/15 are:

Parameter	Yasso07	Yasso20	Change	Effect
p_H	0.04	0.0042	\$-\$90%	Much less humus formation
α_H	0.0033	0.0015	\$-\$54%	Slower humus decomposition
α_W	4.30	5.19	\$+\$21%	Faster W-pool turnover
α_E	0.35	0.13	\$-\$63%	Slower E-pool turnover
α_N	0.22	0.10	\$-\$55%	Slower N-pool turnover

The dramatically reduced p_H ($9.5\times$ less humus formation) is the primary driver of Yasso20 predicting \$20% lower total SOC than Yasso07/15 for Finnish forests, despite the compensating effect of slower humus decomposition.

4.4 RothC

Pool structure

Five pools: $\mathbf{C} = [C_{\text{DPM}}, C_{\text{RPM}}, C_{\text{BIO}}, C_{\text{HUM}}, C_{\text{IOM}}]^T$, representing Decomposable Plant Material, Resistant Plant Material, Microbial Biomass, Humified Organic Matter, and Inert Organic Matter.

Dynamics

RothC operates on a **monthly timestep** $\Delta t = 1/12, \text{yr}$. For each active pool $i \in \{\text{DPM}, \text{RPM}, \text{BIO}, \text{HUM}\}$:

$$C_i(t + \Delta t) = C_i(t) \cdot \exp(-k_i \cdot r(t) \cdot \Delta t) + I_i \cdot \Delta t + \text{transfers}_i$$

where $r(t) = \xi_T(t) \cdot \xi_M(t) \cdot \xi_V$ is the composite rate modifier (normalised so that $\xi_T/47.9$ gives the fractional effect).

The IOM pool is inert: $C_{\text{IOM}}(t) = 0.049 \cdot (\% \text{clay})^{1.139}$.

Decomposed material from each pool:

$$D_i = C_i \cdot (1 - \exp(-k_i \cdot r \cdot \Delta t))$$

This is partitioned into three fluxes governed by the clay-dependent ratio x :

$$x = 1.67(1.85 + 1.60 \exp(-0.0786 \cdot \% \text{clay}))$$

$$\begin{aligned}
f_{\text{CO}_2} &= x/(x+1) \\
f_{\text{BIO}} &= 0.46/(x+1) \\
f_{\text{HUM}} &= 0.54/(x+1)
\end{aligned}$$

So the transfers are:

$$\begin{aligned}
\text{CO}_2 &= \sum_i D_i \cdot f_{\text{CO}_2} \\
C_{\text{BIO}} &+= \sum_i D_i \cdot f_{\text{BIO}} \\
C_{\text{HUM}} &+= \sum_i D_i \cdot f_{\text{HUM}}
\end{aligned}$$

Only DPM and RPM receive external inputs: I_{DPM} , I_{RPM} (partitioned from total litter via the DPM/RPM ratio).

Environmental modifiers

Temperature (monthly):

$$\xi_T = \frac{47.9}{1 + \exp(106 / (T + 18.3))}$$

Moisture — based on accumulated topsoil moisture deficit (TSMD):

$$\text{TSMD}_m = \max(0, E_m - P_m)$$

The moisture modifier reduces decomposition when soil dries:

$$\xi_M = \max(0.2, 1 - \text{TSMD}/\text{TSMD}_{\max})$$

where $\text{TSMD}_{\max}$ depends on clay content.

Vegetation cover:

$$\xi_V = \begin{cases} 0.6 & \text{vegetated (all forest plots)} \\ 1.0 & \text{bare soil} \end{cases}$$

Analytical steady state

The system can be expressed as $\mathbf{A} \mathbf{C}^* = -\mathbf{b}$ for the four active pools, where $\mathbf{A}$ is the 4×4 matrix encoding losses and transfers, and $\mathbf{b} = [I_{\text{DPM}}, I_{\text{RPM}}, 0, 0]^T$.

$$\mathbf{C}_{\text{active}}^* = -\mathbf{A}^{-1} \mathbf{b}$$

The full steady state appends C_{IOM} calculated from clay content.

Default parameters

Parameter	Value	Description
k_{DPM}	10.0	DPM decomposition rate (yr^{-1})
k_{RPM}	0.3	RPM decomposition rate (yr^{-1})

Parameter	Value	Description
k_{BIO}	0.66	BIO decomposition rate (yr^{-1})
k_{HUM}	0.02	HUM decomposition rate (yr^{-1})

4.5 Q-model

Pool structure

Unlike the discrete-pool models above, the Q-model (Bosatta & Ågren, 1985) describes decomposition along a **continuous quality axis** q . Organic matter is characterised by its current quality $q \in (0, q_0]$, where q_0 is the initial quality at litter input. As material decomposes, quality declines and decomposition slows — encoding the universal observation that fresh litter decomposes faster than aged organic matter.

In practice, the model is discretised into **cohort tracking**: each annual litter input forms a cohort whose remaining mass is tracked as a function of age and tissue type.

Five tissue classes are tracked separately, distinguished by initial quality and entry dynamics:

Tissue class	Quality parameter	Entry dynamics
Foliage (needles)	q_{0n} (species-specific)	Immediate
Fine roots	q_{0f} (species-specific)	Immediate
Understorey	q_{0n} (as foliage)	Immediate
Branches	q_{0w} (species-specific)	Gradual fragmentation ($t_{\max,b} = 13$ yr)
Stems	q_{0w} (species-specific)	Gradual fragmentation ($t_{\max,s} = 60$ yr)

Core equations

Microbial uptake rate as a function of quality:

$$u(q) = u_0 \cdot q^\beta$$

where u_0 is the temperature-dependent base rate and β is the quality-rate exponent (default $\beta = 7$).

Temperature dependence (linear form):

$$u_0(T) = u_{00} + u_{01} \cdot T$$

where T is the mean annual temperature (°C), $u_{00} = 0.0855$, and $u_{01} = 0.0157$.

An alternative thermodynamic (Arrhenius) formulation (not implemented here) exists (Bosatta & Ågren 1999):

$$u(q, T) = u_0^{\text{ref}} \cdot \exp\left(\frac{|\Delta G^0|}{Rq} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T}\right)\right)$$

where T is in Kelvin, $R = 8.314, \text{J, mol}^{-1}, \text{K}^{-1}$, and ΔG^0 is the free energy change per enzymatic step. This formulation implies that low-quality substrates have stronger temperature sensitivity than high-quality substrates — a feature not captured by the linear form.

Derived parameters. The quality decline exponent z controls how decomposition slows over time:

$$z = \frac{1 - e_0}{\beta \cdot \eta_{11} \cdot e_0}$$

where e_0 is the initial microbial growth efficiency (0.25) and η_{11} is the carbon use efficiency parameter (0.36). With default parameters, $z \approx 1.19$.

The composite decomposition rate α for a given tissue type combines all factors:

$$\alpha = f_C \cdot \beta \cdot \eta_{11} \cdot u_0 \cdot q_0^\beta$$

where $f_C = 0.5$ is the carbon fraction of organic matter.

Remaining fraction functions

Simple tissues (foliage, fine roots, understorey) enter the soil immediately and follow power-law decay:

$$g(t) = \frac{1}{(1 + \alpha t)^z}$$

Branches fragment gradually over $t_{\max,b}$ years. The remaining fraction integrates decomposition over the fragmenting material. Defining $i_b = \min(t, t_{\max,b})$:

$$g_b(t) = \frac{2 \left[(1 + \alpha t)^{1-z} - (1 + \alpha(t - i_b))^{1-z} (1 - i_b/t_{\max,b}) \right]}{\alpha (1 - z) t_{\max,b}} + \frac{2 \left[(1 + \alpha(t - i_b))^{2-z} - (1 + \alpha t)^{2-z} \right]}{\alpha^2 (1 - z)(2 - z) t_{\max,b}^2} + \left(1 - \frac{i_b}{t_{\max,b}} \right)^2$$

Stems follow the same structure with $t_{\max,s}$ replacing $t_{\max,b}$.

Quality evolution

Quality declines with mass loss:

$$q(t) = q_0 \left(\frac{C(t)}{C(0)} \right)^{1/z}$$

This couples quality and mass: as carbon is lost, the remaining material becomes progressively more recalcitrant.

Transient temperature implementation

The standard Q-model analytical solution assumes constant α (i.e. constant temperature). Under transient climate, we track **cumulative decomposition potential** for each cohort:

$$\tilde{\alpha}(t) = \sum_{\tau=1}^t \alpha(T(\tau))$$

where $\alpha(T(\tau))$ is recalculated each year from the current temperature. The remaining fraction then becomes:

$$g(t) = \frac{1}{(1 + \tilde{\alpha}(t))^{\bar{z}}}$$

This ensures all carbon pools respond to current temperature conditions rather than being locked to the temperature at the time of input — a critical requirement for climate change projections. Without this correction, the model underestimates SOC loss under warming by 20–40%.

Initialisation: hybrid approach

Rather than running a 1500-year numerical spinup, we use an analytical-hybrid approach:

1. **Analytical steady state.** For simple tissues:

$$C_{ss} = L \cdot \sum_{t=1}^{T_{\max}} g(t)$$

where L is the annual litter input. Branches and stems use numerical integration of their respective g_b , g_s functions over 500–1000 years.

2. **Bulk pool initialisation.** The total C_{ss} is treated as a single bulk pool that decomposes via:

$$C_{init}(t) = C_{ss} \cdot (1 + \alpha_0 t)^{1-z}$$

3. **Short spinup** (10–20 years) builds up explicit cohort matrices while the bulk pool decays. Total SOC = bulk pool + new cohorts.

This reduces initialisation time from \$ \$1500 years to 10–20 years while maintaining accuracy.

Default parameters

Parameter	Value	Description
β	7	Quality–rate exponent
e_0	0.25	Initial microbial growth efficiency
η_{11}	0.36	Carbon use efficiency parameter
f_C	0.5	Carbon fraction of organic matter
u_{00}	0.0855	Base uptake rate at 0°C
u_{01}	0.0157	Temperature sensitivity of uptake
$t_{max,b}$	13	Branch fragmentation time (yr)
$t_{max,s}$	60	Stem fragmentation time (yr)
q_{0n}	1.01 (spruce)	Initial quality, foliage
q_{0f}	1.036	Initial quality, fine roots
q_{0w}	1.00	Initial quality, woody tissues

4.6 Common Interface and Implementation Architecture

All five decomposition models share a common calling interface, enabling the simulation loop and Bayesian calibration framework to treat them interchangeably. The implementation is split across two languages: performance-critical numerical code in **Fortran**, and data handling, calibration logic, and input preparation in **R**.

Fortran–R boundary

The guiding principle is to minimise R↔Fortran call overhead during MCMC sampling. Since the calibration loop evaluates the likelihood millions of times across \$ \$3,700 plots, each evaluation should require a **single Fortran call per plot** that runs the full time-series simulation, rather than hundreds of per-timestep round-trips.

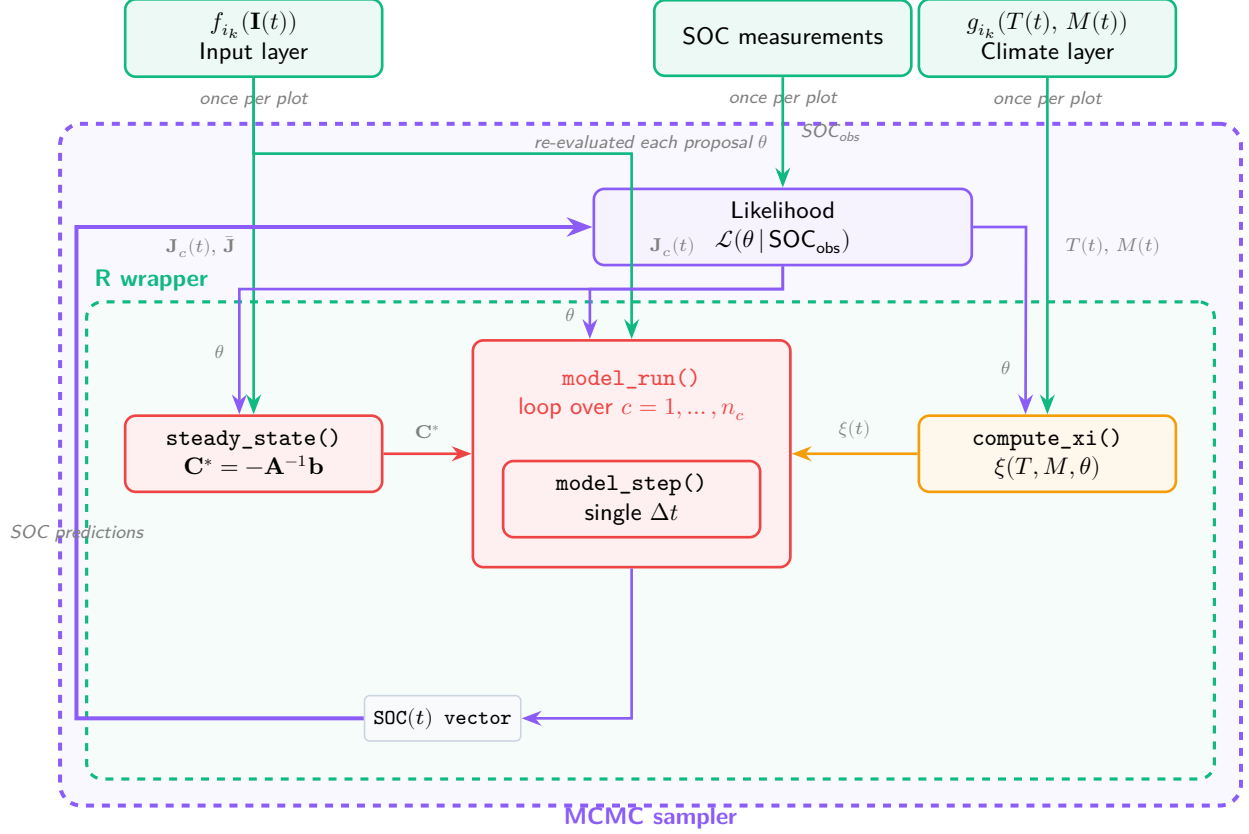


Figure 2: Revised implementation architecture. Three blocks outside the MCMC loop are computed once per plot: the input layer f_{i_k} maps raw litter data to model-specific arrays $\mathbf{J}_c(t)$ and mean inputs $\bar{\mathbf{J}}$; the climate layer g_{i_k} aggregates raw climate to the resolution required by each model; and SOC observations feed the likelihood. At each MCMC proposal θ , the likelihood dispatches parameters to the **R wrapper**, which runs three components: `steady_state()` (Fortran, red) receives θ and $\bar{\mathbf{J}}$ and initialises pools $\mathbf{C}^*$; `compute_xi()` (R, gold) receives θ and the pre-computed climate series to produce $\xi(t)$; and `model_run()` (Fortran, red) receives $\mathbf{C}^*$, $\xi(t)$, θ , and $\mathbf{J}_c(t)$, iterates `model_step()` over all cohorts and timesteps, and returns the SOC trajectory for likelihood evaluation.

Fortran module structure

Each model is implemented as a Fortran module exposing three subroutines. All share the same calling conventions.

model_run() — the top-level entry point called from R via `.Fortran()`. It receives the full pre-processed input arrays, parameter vector, and climate time series for a single plot, and returns the complete SOC trajectory.

```
# R side: single call per plot
result <- .Fortran("yasso15_run",
  n_years   = as.integer(n_years),
  params    = as.double(param_vector), # calibrated parameters
  J_t       = as.double(input_matrix),  # [n_years x n_input_pools]
  T_monthly = as.double(temp_matrix),   # [n_years x 12]
  P_monthly = as.double(precip_matrix),  # [n_years x 12]
  diameter  = as.double(wood_diameters), # [n_litter_types]
  C_out     = double(n_years * n_pools), # output: [n_years x n_pools]
```

```
resp_out = double(n_years)           # output: annual respiration
)
```

Internally, `model_run()` performs:

1. Call `model_steady_state()` to compute $\mathbf{C}^*$ from mean climate and inputs.
2. Initialise pools from $\mathbf{C}^*$ (or from the hybrid approach for Q-model).
3. Loop over years (and months for RothC), calling `compute_xi()` and `model_step()` at each timestep.
4. Write annual pool states and respiration to output arrays.

`model_steady_state()` — computes the analytical equilibrium. For Yasso and RothC, this is a matrix inversion $\mathbf{C}^* = -\mathbf{A}^{-1}\mathbf{b}$. For the Q-model, it performs the numerical summation of remaining fractions over a truncated age series.

`model_step()` — advances the pool state by one timestep Δt :

Model	Δt	Core operation
Yasso07/15/20	1 yr	Build $\mathbf{A}$, compute matrix exponential $e^{\mathbf{A}}$, update $\mathbf{C}$
RothC	1/12 yr	Exponential decay per pool, partition decomposed $\mathbf{C}$ via clay ratio x
Q-model	1 yr	Update cumulative $\tilde{\alpha}$ per cohort, compute remaining fraction, prune old cohorts

`compute_xi()` — evaluates the environmental modifiers from climate data and static site properties. Called internally by `model_step()` (or once per year for annual models, 12 times for RothC).

R wrapper layer

The R wrapper for each model handles:

- **Climate mapping:** calls `map_climate_<model>()` once per plot to produce the climate data frame expected by that model. Yasso07/15 use `map_climate_yasso07()` (annual mean + amplitude); Yasso20 and RothC share `map_climate_monthly()` (full monthly series).
- **Input mapping:** calls `map_inputs_<model>()` once per plot to aggregate the six raw cohorts into model-specific litter arrays. Yasso07, Yasso15 and Yasso20 share the same input structure via `map_inputs_yasso07()`.
- **Parameter mapping:** translates the named BayesianTools parameter vector into the positional double-precision vector expected by Fortran.
- **Output formatting:** reshapes the flat Fortran output arrays into named R structures for likelihood evaluation and diagnostics.

All models are run plot-by-plot with a consistent `lapply` $\rightarrow$ `rbind` pattern. Climate and input mapping are performed once per plot, outside the MCMC loop:

```
# --- Once, before calibration ---
plots      <- unique(input_raw$plot_id)
Yasso07_climate <- map_climate_yasso07(input_raw) # annual: mean, amplitude, precip
monthly_climate <- map_climate_monthly(input_raw) # monthly: shared by Yasso20 + RothC
Yasso07_inputs  <- map_inputs_yasso07(input_raw) # shared by Yasso07, Yasso15, Yasso20
RothC_inputs    <- map_inputs_rothc(input_raw, site_raw)

# --- Per-plot simulation (inside likelihood, called each MCMC proposal) ---
result <- do.call(rbind, lapply(plots, function(pid) {

  clim  <- Yasso07_climate[Yasso07_climate$plot_id == pid, ]
```



```

inputs <- Yasso07_inputs[Yasso07_inputs$plot_id == pid, ]

# Stage 1: Xi time series from plot-specific climate
xi <- compute_xi_yasso07(clim, params)

# Stage 2: Steady-state initialisation from plot-specific means
C_init <- yasso07_steady_state(
  params = params,
  nwl_mean = colMeans(inputs[, c("nwl_A", "nwl_W", "nwl_E", "nwl_N")]),
  fwl_mean = colMeans(inputs[, c("fwl_A", "fwl_W", "fwl_E", "fwl_N")]),
  cwl_mean = colMeans(inputs[, c("cwl_A", "cwl_W", "cwl_E", "cwl_N")]),
  xi_mean = mean(xi)
)

# Stage 3: Transient simulation - single Fortran call per plot
out <- yasso07_run(inputs, params, C_init, xi)

cbind(plot_id = pid, out)
}))

```

What stays in R vs. Fortran

Component	Language	Rationale
Climate mapping (<code>map_climate_*</code>)	R	Called once per plot; benefits from R data wrangling
Input mapping (<code>map_inputs_*</code>)	R	Called once per plot; benefits from R data wrangling
ξ_k — modifier computation	Fortran	Called millions of times inside MCMC loop
f_{d_k} — decomposition step	Fortran	Innermost loop; matrix operations
Steady-state solver	Fortran	Matrix inversion, numerical integration
Full simulation loop	Fortran	Avoids per-timestep R↔Fortran overhead
Parameter mapping	R	String/name handling; negligible cost
Likelihood evaluation	R	Interface with BayesianTools; simple arithmetic
I/O, aggregation, diagnostics	R	Flexibility; not performance-critical

4.7 Yasso Family: Three-Stage Call Pattern

All three Yasso versions (07, 15, 20) follow the same three-stage execution pattern in R. The differences are confined to Stage 1 (Xi computation) and the climate object passed to each stage.

Climate and input mapping functions

Model	Climate function	Input function
Yasso07	<code>map_climate_yasso07()</code> → annual mean, amplitude, precip	<code>map_inputs_yasso07()</code>
Yasso15	<code>map_climate_yasso07()</code> (reused)	<code>map_inputs_yasso07()</code> (reused)
Yasso20	<code>map_climate_monthly()</code> (shared with RothC)	<code>map_inputs_yasso07()</code> (reused)

All three versions share the same litter input structure: six raw cohorts aggregated to three litter types (nwl, fwl, cwl) with AWEN fractions preserved.

Stage 1 — Xi: climate modifier time series

The key structural difference among the three versions is in how ξ is computed:

Model	Function	Climate resolution	Pool-group specificity
Yasso07	<code>compute_xi_yasso07()</code>	Annual (mean + amplitude → 4-point Gaussian)	Single ξ for all pools
Yasso15	<code>compute_xi_yasso15()</code>	Annual (mean + amplitude → 4-point Gaussian)	Three ξ : AWE, N, H
Yasso20	<code>compute_xi_yasso20()</code>	Monthly (exact 12-month average)	Three ξ : AWE, N, H

Stage 2 — Steady state: initial pool conditions C*

Steady state is computed from plot-specific mean climate and mean litter inputs. For Yasso20, a synthetic 12-row `mean_clim` data frame is constructed from the plot's monthly climatology:

```
# Yasso07 / Yasso15
C_init <- yassoXX_steady_state(
  params = params,
  nwl_mean = colMeans(inputs[, c("nwl_A", "nwl_W", "nwl_E", "nwl_N")]),
  fwl_mean = colMeans(inputs[, c("fwl_A", "fwl_W", "fwl_E", "fwl_N")]),
  cwl_mean = colMeans(inputs[, c("cwl_A", "cwl_W", "cwl_E", "cwl_N")]),
  xi_mean = mean(xi) # or list(xi_awe, xi_n, xi_h) for Yasso15
)

# Yasso20 - requires a synthetic single-year mean climate data frame
# precip must be monthly mean (NOT divided by 12); the function sums
# internally over 12 months to recover the annual total
mean_clim <- data.frame(
  year = 9999L,
  month = 1:12,
  temp_air = tapply(clim$temp_air, clim$month, mean, na.rm = TRUE),
  precip = mean(clim$precip) # monthly mean ~50 mm; sums to ~600 mm/yr internally
)
C_init <- yasso20_steady_state(mean_clim, params, nwl_mean, fwl_mean, cwl_mean)
```

Stage 3 — Transient simulation

All three versions return a flat data frame with columns `year | A | W | E | N | H | total_soc | respiration`.

```
out_07 <- yasso07_run(inputs_df,    params, C_init, xi_array)
out_15 <- yasso15_run(inputs_df,    params, C_init, xi_arrays, precip, leac)
out_20 <- yasso20_run(climate_df,   inputs_df, params, C_init, leac)
#                                     ~~~~~~
#                                     monthly_climate - xi_arrays and precip derived internally
```